Ji 2020 cutaneous SCC — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

8/10 samples form one continuous, sample-mixed manifold (iLISI≈4.0; single kNN component; 13 detached cells removed). Cell-cycle regression required.

Gate 2 — module geography → GO (branching)

A branching wishbone: three lobes (Basal, Differentiated, Mixed) meet at one convergence with a smooth monotone pseudotime (r≈0.96). Rooted at the Basal tip (COL17A1⁺/TP63⁺ progenitor).

Gate 3 — cross-repression → GO

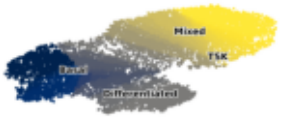
On the Basal root population (3,524 cells), Gast↔pSom gene loadings mutually anti-correlate: r = −0.41 (permutation p≈0.0005) — within range of branching anchors (Puram −0.49, Filbin −0.34).

STEP FIGURES

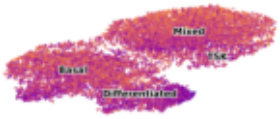
Manifold — samples



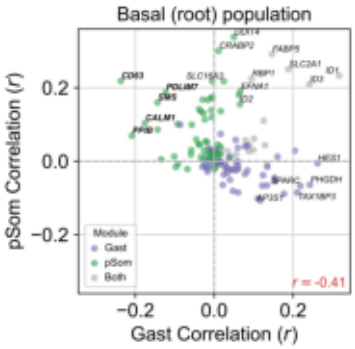
Trajectory / root



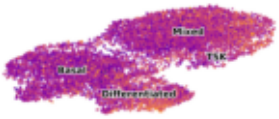
Gastrulation



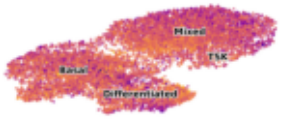
Gate 3: r = −0.41



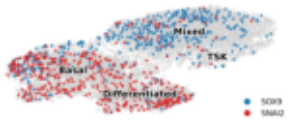
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Cutaneous SCC (Ji et al., 2020), 10x — 10 patients, 48,164 cells; 10,598 malignant → 10,518 after QC. Subtypes Mixed / Basal / Differentiated / TSK. Branching (hub-and-spoke wishbone).

Verdict.
Branching — clean 3-tip wishbone rooted at the Basal progenitor tip, competitive Gast↔pSom cross-repression (r=−0.41, p≈0.0005) at the Basal root; Crest terminal. All three gates GO. EMT-TFs SOX9 / SNAI2.

Parameter choices

Min-cell threshold	≥100 (gap 40→268; keeps 8/10)	**Cell-cycle regression**	S + G2M regressed — key to clean manifold
Integration	Harmony on sample, θ=2; n_pcs=20	**Embedding**	UMAP min_dist=0.4, spread=1.0
Trajectory	scFates PPT N=40, λ=700, σ=0.8 (3-tip wishbone)	**Root**	Basal tip (COL17A1⁺/TP63⁺ progenitor)
Gate-3 pop / r	Basal root (3,524 cells); r=−0.41, p≈0.0005	**EMT-TFs shown**	SOX9 (basal) / SNAI2 (Differentiated arm)